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Consumer perceived value: The development of a multiple item scale

Jillian C. Sweeney[a,*], Geoffrey N. Soutar[b]

[a]*Faculty of Economics and Commerce, University of Western Australia, Perth, WA 6009, Australia*
[b]*Graduate School of Management, University of Western Australia, Perth, WA 6009, Australia*

Abstract

Value creation is widely discussed in the practitioner literature and is often a part of organizations' mission statements and objectives. It is seen by many commentators as the key to long-term success, with Albrecht (1992, p 7) arguing that "the only thing that matters in the new world of quality is delivering customer value." Despite this emphasis, little research has addressed the value construct itself and there is no well-accepted value measure, even in the retail environment in which customers evaluate products before purchase.

The present research project describes the development of a 19-item measure, PERVAL, that can be used to assess customers' perceptions of the value of a consumer durable good at a brand level. The measure was developed for use in a retail purchase situation to determine what consumption values drive purchase attitude and behavior. Four distinct, value dimensions emerged that were termed emotional, social, quality/performance and price/value for money. The reliability and validity of the scale was assessed in a prepurchase situation, using exploratory and confirmatory analyses. All four value dimensions were found to help significantly in explaining attitudes and behavior. The scale was also tested in a postpurchase situation and found to be both reliable and valid in this context as well. The PERVAL scale has a variety of potential applications and can serve as a framework for further empirical research in this important area. © 2001 by New York University. All rights reserved.

1. Introduction

Perceived value, a strategic imperative for producers and retailers in the 1990s, will be of continuing importance into the twenty-first century (Vantrappen, 1992; Woodruff,

* Corresponding author. Tel.: +61-9-380-1438; fax: +61-9-380-1004.
E-mail addresses: jsweeney@ecel.uwa.edu.au (J.C. Sweeney), gsoutar@ecel.uwa.edu.au (G.N. Soutar).

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1997; Forester, 1999). Indeed, from a retailing perspective, Hartnett (1998, p 21) noted that "when [retailers] satisfy people-based needs, they are delivering value, which puts them in a much stronger position in the long term," while Burden (1998, p 2) commented that "successful retailers increasingly target their offers towards two consumer categories: those with an emphasis on value and those for whom time pressure is the key." This move to value in retailing seems to be a global phenomenon as "the most compelling [Asian retail] opportunities are at the value end of the market given that consumers in Asia today are . . . much more value conscious than they were in the mid-1990s" (Treadgold, 1999, p 45).

If it is true that retail customers are "value-driven" (Levy, 1999), then managers need to understand what customer's value and where they should focus their attention to achieve this needed market place advantage (Woodruff, 1997). Despite value's importance, however, there has been relatively little empirical research to develop an in-depth understanding of the concept. Even less research has focused on specifying its domain or on developing a practical and operational perceived value scale. The present paper outlines the development of such a scale and begins with a brief discussion of the origins of the consumption value construct before discussing the present study.

Zeithaml (1988, p 14) has suggested that perceived value can be regarded as a "consumer's overall assessment of the utility of a product (or service) based on perceptions of what is received and what is given." She referred to this assessment as a comparison of a product or service's 'get' and 'give' components. The most common such definition of value is the ratio or trade-off between quality and price (e.g., Chain Store Age, 1985; Cravens, Holland, Lamb & Moncrieff, 1988; Monroe, 1990), which is a value-for-money conceptualization.

Clearly, these two components (quality and price) have different and differential effects on perceived value for money. Zeithaml (1988) argued that some consumers perceive value when there is a low price, others perceive value when there is a balance between quality and price. Thus, for different consumers, the components of perceived value might be differentially weighted. Additionally, Zeithaml (1988) found that some consumers obtained value from all relevant 'get' and 'give' components, leading to her definition of perceived value.

Other authors have also suggested that viewing value as a trade-off between only quality and price is too simplistic (e.g., Schechter, 1984, Bolton & Drew, 1991). Porter (1990, p 37), for example, talked about providing "superior value to the buyer in terms of product quality, special features, or after-sale service." These views suggest that existing value constructs are too narrow and that dimensions other than price and quality would increase the construct's usefulness. A more sophisticated measure is needed to understand how consumers value products and services and the present study was an attempt to create such a measure.

The following section of this article outlines a conceptual framework for the suggested measure while subsequent sections discuss the process through which the scale was developed and tested while the final section note some of the study's theoretical and managerial implications.

2. A conceptual framework

Consumer research has evolved from a focus on the cognitive aspects of decision making to include intrinsic aspects, so that an object or experience can be seen to be valued for its own sake. Holbrook and Hirschman (1982), for example, argued for an experiential perspective that included the symbolic, hedonic and esthetic aspects of the consumption process. They suggested that the existing information processing perspective implied products were largely judged through utilitarian criteria, based on how well a product or service serves its intended purpose or performs its proper function. An experiential perspective views products or services through hedonic criteria, based on an appreciation of the good or service for its own sake. Other researchers (e.g., Batra & Ahtola, 1990) supported the presence of distinct utilitarian and hedonic components, which have been referred to as 'thinking and feeling' dimensions.

In particular, Babin, Darden and Griffin (1994) developed a specific measure of shopping value that includes utilitarian and hedonic components, while Richins (1994) created a 'possession rating scale.' While her scale included utilitarian and hedonic components, it related to possessions people already own. Indeed, many of the possessions examined were sentimental objects, such as photograph albums. This is in contrast to the present study, which was concerned with measuring perceptions of value of consumer durable goods prior to, as well as soon after, purchase as an aid to understanding consumers' decision processes and choice behavior.

A broader theoretical framework of perceived value was developed by Sheth, Newman and Gross (1991a, 1991b), who regarded consumer choice as a function of multiple 'consumption value' dimensions and that these dimensions make varying contributions in different choice situations. They suggested five dimensions (social, emotional, functional, epistemic and conditional value), relating specifically to the perceived utility of a choice, whether at the decision to buy level (buy or not buy), at the product level (product type A or product type B) or at brand level (brand A or brand B). Their work provides the best foundation for extending existing value constructs as it was validated through an intensive investigation of the variety of fields in which value has been discussed, including economics and social and clinical psychology.

Functional value was seen to be the key influence on consumer choice. However, Sheth et al. (1991b) found the other value dimensions were also influential in some situations. For example, while functional and social value dominated the decision as to whether to use filtered or unfiltered cigarettes, emotional value was key to the decision to smoke. It is clear that different value dimensions may be important depending on the decision level (e.g., buy/not buy or buy brand A/brand B), as well as on the type of product or service being considered.

2.1. Value dimensions are inter-related

Sheth et al. (1991a, p 12) argued that value dimensions are independent as they "relate additively and contribute incrementally to choice." However, prior research suggests that the hedonic and utilitarian components of attitude may be related (e.g., Osgood, Suci & Tannenbaum, 1957). For example, the purchase of an attractive carpet is likely to increase the chances of a favorable emotional as well as a favorable functional response. Conse-

quently, value dimensions may not be independent. Indeed, many other multidimensional constructs, including organizational commitment (Mowday, Steers & Porter, 1979), well-being at work (Warr, 1990), retail service quality (Dabholkar, Thorpe & Rentz, 1996) and communication-evoked mental imagery (Babin & Burns, 1998), have been found to have separate but correlated dimensions. Consequently, the value dimensions developed in the present study were allowed to be interrelated.

2.2. Perceived value is different from satisfaction

Perceived value has been widely discussed at a generic level (e.g., *providing value*), particularly in the practitioner literature and can easily be confused with satisfaction (e.g., *meeting customers' needs*). However these constructs are distinct. While perceived value occurs at various stages of the purchase process, including the prepurchase stage (Woodruff, 1997), satisfaction is universally agreed to be a postpurchase and postuse evaluation (e.g., Hunt, 1977; Oliver, 1981). As a consequence, value perceptions can be generated without the product or service being bought or used, while satisfaction depends on experience of having used the product or service. In addition, satisfaction has been conceptualized as a unidimensional construct, largely due to the assumption that it varies along a hedonic continuum from unfavorable to favorable (Westbrook & Oliver, 1991) and to its conceptualization as a consequence, outcome or summary variable in comparison to value, which is antecedent to it (Parasuraman, 1997). In contrast, we conceptualize value as a multidimensional constructs.

2.3. The proposed scale

As mentioned above, Sheth et al.'s (1991a, 1991b) model provides a strong foundation from which to build a perceived value scale. However, these authors argued that functional value was created by attributes such as reliability, durability and price. The first two of these attributes have often been seen as aspects of quality and, in other value models, quality and price are held to have separate influences on perceived value; quality having a positive and price a negative effect (e.g., Dodds, Monroe & Grewal, 1991). Consequently, it could be argued that price and quality are functional subfactors that contribute separately to perceived value and that they should be measured separately.

We began our study, therefore, by including potentially separate price and quality dimensions in the value construct. The aim was to develop a useful, parsimonious and practical scale that could be easily applied in a variety of purchase situations. The initial stages of the development of such a perceived value scale (called PERVAL) are discussed in the following section.

3. The scale development process

The evidence already discussed suggests that there are distinct aspects of consumption value. The present section describes the process used to establish the content for these dimensions and to validate the scale psychometrically and theoretically. The process follows

Churchill's (1979) approach for developing measures of multiple-item marketing constructs. After the development of an initial set of items, two scale purification stages were undertaken. While the first stage used a student sample, the second stage used a more diverse sample of consumers.

3.1. The development of an initial set of items

In the first phase of the research we explored the ideas and opinions that consumers held about *consumption value.* Six focus groups were conducted among adult consumers in a major Australian city. Ten people attended each session. Respondents, balanced between male and female, were from a range of occupations such as white and blue collar, home duties and retired people. They were aged from 25 to 59 years.

This diversity of focus group members was planned to represent buyers of the products discussed in the various sessions with the key purpose of the sessions to generate items measuring consumption values. A different product was used in each session to stimulate participant thinking in different directions. The goods used included clothing and durable goods. During the trial group interview, it became clear that asking consumers why they valued a particular brand, such as Panasonic, when considering a purchase evoked a greater range of value items than asking why they valued a certain product type, such as stereo or mono television. For this reason consumers were asked for their opinions about the values associated with brands, rather than product types.

Group members were asked a series of questions to provoke thought about consumption value. Items generated tended to be attributes of the brand. For example, *color, looks, comfort, features, price* and *reliability* were mentioned. Following this, respondents were asked why these aspects were important to them in an attempt to better understand the underlying benefits the "aspects" provided since benefits would be more useful in developing generic consumption value items.

As a consequence of this exploratory phase, it was found that many of the items produced were *generic.* After taking account of identical or equivalent items, a total of one hundred and seven consumption value statements were retained for further evaluation.

Nine academic colleagues evaluated the items obtained from the focus groups to ensure they were representative of the scale's domains. To assist, we gave each judge a description of each of the six value dimensions based on Sheth et al.'s (1991a) discussion and included the two suggested functional subdimensions of quality and price. The use of experts as judges of a scale's domain has been commonly used in marketing (e.g., Zaichowsky, 1985; Babin & Burns, 1998).

Items that seven or more of the nine judges classified as representative of a specific value dimension were kept for further scale development. This resulted in ninety-two items being retained from the one hundred and seven item originally assessed. The high proportion retained suggests that the experts had little difficulty in classifying the various items into common value dimensions. Of interest, all judges categorized the item "this product offers value for money," which has sometimes been used as a single item perceived value measure, into the price dimension.

None of the items generated were judged to reflect epistemic value, which relates to the

surprise or novelty aspect of a product, or to conditional value, which refers to the conditional effects of a specific situation on value perceptions. Epistemic value relates to a product's capacity to arouse curiosity, offer novelty or satisfy a desire for knowledge and may be important for consumers who are considering new experiences. This is particularly likely in the case of experiential services such as holidays, adventures or even shopping trips (Sheth et al., 1991a; Babin, Darden & Griffin, 1994) and is possibly less important when considering the purchase of a durable good.

Sheth et al. (1991a) described conditional value as being derived from temporary functional or social value, hence it is arises when situational factors, such as an illness or some specific social situation, moderate the perceived value-outcome process. Hence conditional value can be described as a specific case of other types of value. Conditional value was seen as less critical in the present study as the aim was to develop a general value measure. Consequently these two aspects were not included and the perceived value scale was based on four dimensions, quality, price, emotional value and social value, with quality and price being seen as subcomponents of functional value, as suggested earlier.

Following a pilot survey of the accepted items with twelve students, seven items were eliminated as they were seen to be too product specific or repetitive. In all, 34 functional items (17 quality, 15 price), 29 social and 22 emotional items were retained as the initial basis for a perceived value scale.

One of the prime considerations in scale development is the adequacy with which a specified domain of content is sampled. The focus group approach used to generate items and the variety of goods discussed in these groups suggests that the scale has content validity.

3.2. Data collection, stage one

We used an initial quantitative procedure to reduce the number of items and to examine the resulting scale's psychometric properties. Specifically, in the first two stages, value was examined in an in-store prepurchase situation. A total of 273 third year or postgraduate students at three Australian universities participated in the first quantitative stage of the research by responding to a questionnaire containing the total set of 85 items. Each student was asked to recall a situation in a shop in the last three months when they had looked at a particular durable product, which they could identify by brand and price, but which they had not bought. The product could have been an item that were thinking of buying or something that they had no intention of buying, either because they couldn't afford it, didn't need it or disliked it. The variation in behavioral intentions regarding the product was important, since it was expected that a similar variation in perceived value would result, increasing the ability to properly test the scale. Four weeks later an identical 'retest' questionnaire, in which students were asked to re-evaluate the same product, was administered to 130 of these students.

To enable an assessment of convergent, discriminant and criterion related validity of the constructs, respondents were asked to answer a series of additional items derived from the literature during both stages. For example, two items selected to represent social value were adapted from Bearden, Netemeyer, and Teel's (1989) 'Consumer Susceptibility to Interper-

sonal Influence' scale. Respondents evaluated all items on a seven-point Likert type scale, ranging from strongly disagree (1) to strongly agree (7).

3.3. Item reduction and exploratory investigation of dimensionality, stage one

Seventeen items were deleted from the scale as they had test-retest correlations of less than 0.50, suggesting they generated more noise than information. During the *retest* students were also asked to state whether any of the items were inappropriate or irrelevant. We asked this to overcome potential irritation or social reaction caused by some of the social value statements, such as "would make people appreciate me more." Five items from the social and emotional dimensions were deleted for this reason. In all, 63 items were retained that represented aspects of functional, emotional and social value.

Following this outcome and, as recommended by Churchill (1979), we further reduced the scale by investigating coefficient alpha and plotting the item-to-total scale correlations for each dimension. Items that produced a sharp drop in the plotted pattern were eliminated. After this, 33 items remained.

To this point, we had not checked for a possible overlap of items across dimensions. With this in mind, we undertook a four-dimensional principal components analysis with varimax rotation. A clear factor pattern emerged. After the iterative deletion of a small number of items that had their highest loading on an incorrect factor or an almost equal loading on more than one factor, 24 items remained. A further exploratory principal components analysis was undertaken on this reduced item set that revealed a clear factor pattern.[1]

3.4. Reliability and validity of scale—stage one

Having established the four dimensions of the scale, we conducted a confirmatory factor analysis. As suggested by Bollen (1989), a null model, in which no factors were considered to underlie the observed variables, correlations between observed indicators were zero and the variances of the observed variables were not restricted, was tested against a series of models, namely:

- A one factor model (suggesting that the observed variables represent a single value dimension)
- A three factor model (in which price and quality are suggested to represent a single functional dimension rather than two dimensions, in addition to the emotional and social value dimensions)
- A four factor model (in which the dimensions are as proposed in the earlier discussion)

The results, shown in Table 1, support the proposed four-factor solution, comprising the quality, price, emotional and social value dimensions. Not only did this model have the lowest χ^2 and highest adjusted goodness of fit index, but also the highest noncentrality index (RNI). This index was developed as an unbiased estimator of the Bentler-Bonnett CFI and is recommended for the comparative analysis of models (McDonald & Marsh, 1990; Bagozzi & Heatherton, 1994). Further, the improvements over the three factor, one factor and null models were significant.

TABLE 1
Comparative Analysis of Models of Various Dimensionalities—Stage One

Model	χ^2	DF	AGFI[a]	RNI
Null	2097.23	36	0.25	n.a.
One factor	767.86	27	0.55	0.64
Three factor	242.44	24	0.81	0.89
Four factor	38.14	21	0.96	0.99

Stage Two

Model	χ^2	DF	AGFI[a]	RNI[b]
Null	1990.16	36	0.15	n.a.
One factor	644.40	27	0.42	0.68
Three factor	229.02	24	0.77	0.90
Four factor	24.68	21	0.96	1.00

[a]Adjusted goodness of fit index is denoted by AGFI and the relative non-centrality index by RNI.
[b]RNI = $\{(\chi^2_n\text{-}df_n)\text{-} (\chi^2\text{-}df)\}/(\chi^2_n\text{-}df_n)$- where n is the null model.

The discriminant validity of the four dimensional scale was investigated in two ways. First, the test that the correlation between constructs is significantly less than one was used (Bagozzi & Heatherton, 1994). In practice this test requires an examination of the confidence interval surrounding the estimate. Should the correlation plus or minus two standard errors include the value one, discriminant validity is not supported. The highest correlation between dimensions was 0.71 (between the emotional and quality scales). The associated confidence interval was 0.65 to 0.77. Hence discriminant validity was supported for all pairs of dimensions.

Second, Fornell and Larcker's (1981) discriminant validity test was conducted. This test requires that, when taking any pair of constructs, the average variance extracted for each construct should be greater than the squared structural path coefficient between the two constructs. In the present case these requirements were met for all pairs of constructs, with the average variance extracted ranging from 0.71 to 0.77. This exceeded the squared path coefficient in all cases, since the maximum value of the squared path was 0.50. These results support the distinction of the constructs included in the model, even when measurement error is considered. In addition, high levels of reliability were achieved, the reliability of the individual scales ranging from 0.82 to 0.91.

Convergent validity is also supported as the average variance extracted clearly exceeded 0.50 for all dimensions (Fornell & Larcker, 1981). Convergent and discriminant validity were also assessed through an examination of the correlations of the four dimensions, represented by factor scores, with alternative measures of functional, emotional and social value that were included in the questionnaire. This analysis supported the convergent and discriminant validity of the scale. The scale also behaved as expected in that correlations between the dimensions and measures of related constructs, such as likelihood of purchase, were significant and positive, supporting criterion-related validity. Correlations in stages 1 and 2 were similar, hence only the results from stage 2 are shown (Table 4).

These initial results supported the proposed four dimensional model of perceived value. Although these results provided evidence of reliability, construct validity and criterion-related

validity they were based on a student sample. The 24-item instrument was therefore re-examined using an independent and more diverse second data set, as recommended by Churchill (1979).

3.5. Data collection, stage two

A telephone survey was conducted among adults aged 18 and over in the Perth Metropolitan area in Western Australia. Telephone numbers were selected at random and calls made in the evening and weekends to avoid a bias against working adults. The same, in-store prepurchase situation was used as in stage one of the study. Respondents were asked to think of a situation in a shop in the last three months or so, when they had looked at a particular durable product. A wide variety of durable goods were selected by respondents, including clothing, footwear, furniture, cars, computers, sports goods and household appliances. A total of 875 respondents were approached, 210 of these refused to participate, resulting in an effective response rate of 76%. Of the remaining 665, 362 failed to qualify because they could not clearly recall such a purchase situation. This left 303 interviews for use in the second stage of analysis. Two thirds of the sample was female, and almost two thirds were between 18 and 44 years of age. This corresponds to the profiles of shoppers surveyed in previous studies in Western Australia (e.g., Sweeney, Soutar & Johnson, 1999).

The main objective of the second stage was to evaluate the robustness of the 24-item scale intended to measure perceived value. The procedure involved several steps, similar to those used in stage one.

3.6. Scale purification—stage two

The item-to-total correlations were calculated for the four dimensions. As in the first stage, items that created a sudden drop in the plotted item to total scale correlation patterns were dropped. Following this, an exploratory principal components analysis was conducted of the remaining items. At this stage the scale was reduced to 19 items. An exploratory factor analysis of the remaining 19 items revealed a stable structure of four dimensions that can be described as follows:

Emotional value	the utility derived from the feelings or affective states that a product generates
Social value (enhancement of social self-concept)	the utility derived from the product's ability to enhance social self-concept
Functional value (price/value for money)	the utility derived from the product due to the reduction of its perceived short term and longer term costs
Functional value (performance/ quality)	the utility derived from the perceived quality and expected performance of the product

TABLE 2
Summary of Final Results from Exploratory Factor Analysis—Stage Two

Item	Quality	Emotional	Price	Social
has consistent quality	0.82	0.28	0.21	
is well made	0.79	0.31	0.26	0.23
has an acceptable standard of quality	0.76	0.38	0.20	
has poor workmanship (*)	0.76	0.25	0.26	
would *not* last a long time (*)	0.76	0.20		
would perform consistently	0.70	0.31	0.22	
is one that I would enjoy	0.37	0.80		0.28
would make me want to use it	0.32	0.77		0.26
is one that I would feel relaxed about using	0.37	0.76	0.21	
would make me feel good	0.32	0.74	0.21	0.36
would give me pleasure	0.35	0.71		0.33
is reasonably priced			0.90	
offers value for money	0.30		0.82	
is a good product for the price	0.33	0.35	0.76	
would be economical	0.25		0.72	
would help me to feel acceptable				0.83
would improve the way I am perceived				0.83
would make a good impression on other people	0.26	0.29		0.74
would give its owner social approval		0.26		0.60
Eigen value	9.53	2.22	1.47	1.00

(*) reverse scored
Percentage of variance extracted by the four factors was 75%
Note: Loadings of less than 0.20 are not shown to improve readability

In both stages one and two, the item "this product offers value for money" clearly related to the price dimension. Correlations between this item and the other dimensions were extremely low. While the loadings of this single item onto the price factor in exploratory factor analysis were 0.72 and 0.82 in stages one and two respectively, loadings on other dimensions were far lower. The maximum loading of this item on other dimensions was 0.30, on the quality factor in stage 2.

3.7. Reliability and validity of final scale—stage two

The exploratory factor analysis of the 19 items is shown in Table 2. A clear four-factor structure is evident.

Once again, the suggested four-factor scale structure was compared to the three factor, single factor or null models using a confirmatory factor analysis approach (Table 1). Although the χ^2 value for the four factor model was significant, the four factor model created a significant improvement over the three factor, one factor and null models. Other fit indices, in particular the RNI, indicated that the four-factor solution fit the data well.

TABLE 3
Correlations between Constructs following Confirmatory Factor Analysis—Stage Two

	Quality	Emotional	Price	Social
Quality	0.91			
Emotional	0.74 (0.03)	0.94		
Price	0.47 (0.05)	0.68 (0.04)	0.80	
Social	0.57 (0.05)	0.49 (0.05)	0.26 (0.06)	0.82

NB Standard errors appear in brackets, while composite reliabilities appear on the diagonal
Reliability of linear composite of scale (19 items) = 0.96

The reliabilities of the factors and the total scale are shown in Table 3. The reliabilities range from 0.80 to 0.94, while total scale reliability was 0.96. Discriminant validity was again also evident in the stage two results. When using the confidence interval surrounding the correlation between constructs, the highest correlation was again between the quality and emotional factors (0.74), and the corresponding standard error was 0.03 (Table 3). Employing Fornell and Larcker's (1981) test, the average variances extracted ranged from 0.69 to 0.88, while the square of the path between the constructs was a maximum of 0.55.

As in stage one, convergent and discriminant validity were examined using alternative measures of the emotional, functional and social dimensions derived from previous research. It can be seen from Table 4, that the emotional, social and functional (quality-performance) and functional (price) factors have a higher correlation with the respective items representing emotional, social, quality and price aspects than with other items. This supports the convergent and discriminant validity of these four scales.

TABLE 4
Convergent and Discriminant Validity—Stage Two (correlations of factor scores with independent items)

	Factor 1 Quality	Factor 2 Emotional	Factor 3 Price	Factor 4 Social
This item would, in functional terms, perform well	**0.63**	0.36	0.26	0.16
This item would arouse positive feelings in me	0.33	**0.60**	0.05	0.42
I would like this item	0.41	**0.72**	0.21	0.26
This item is too expensive	−0.01	0.08	**−0.63**	−0.08
This item would be approved of by others				
If I bought or used this item, it would create a favorable perception of me among other people	0.21	0.26	−0.02	**0.63**
This item has a positive social image	0.23	0.29	−0.09	**0.55**
I would be willing to buy this item at this store	0.35	0.62	0.38	0.19
I would recommend this item to friends or relatives	0.46	0.48	0.31	0.29
I would not expect any problems with this item	0.55	0.31	0.21	0.21

TABLE 5
Relative Importance of the Four-Value Dimensions in Predicting Behavior: Results of Stepwise Regression

Outcome	Dimension	Standardized Slope coefficient	T values	Adjusted R^2	Variance explained by single item—'This prodcut offers value for money'
I would be willing to buy this item at this store	Quality	0.35	9.20	0.68	0.29
	Emotional	0.62	16.21		
	Price	0.38	9.93		
	Social	0.19	4.88		
I would recommend this item to friends or relatives	Quality	0.46	11.30	0.62	0.23
	Emotional	0.48	11.75		
	Price	0.31	7.53		
	Social	0.29	7.09		
I would not expect any problems with this item	Quality	0.55	11.38	0.48	0.24
	Emotional	0.31	6.49		
	Price	0.21	4.27		
	Social	0.21	4.30		

Whether the measure PERVAL behaves as expected with measures of other constructs external to the scale itself, its criterion-related validity, was assessed by examining its relationship with other conceptually related variables. These included a willingness to buy the product, a willingness to recommend the product and not expecting problems with the product. It would be expected that if a consumer perceived a product to have a high *value,* he or she would be more willing to buy the product, be more willing to recommend the product and expect fewer problems with the product. As can be seen in Table 4, the results support these expectations.

The multiple value dimensions also explained consumer outcomes better than a single 'value-for-money' item, as can be seen in Table 5. The explained variance in three different outcome variables ('willing to buy in this store', 'I would recommend this item to friends or relatives' and 'I would not expect any problems with this item') was between 48 and 68% when the multiple value dimensions were used, but less than half this when the traditional single item of 'value for money' item was used. Using the four dimensions in combination explained considerably more of the variance in possible outcomes than did a single item. It is also apparent that the dimensions had differential effects on the various outcomes. The emotional value dimension was of great importance in predicting willingness to buy in particular, while perceptions of quality had a particular influence on people's expectations of problems.

Overall, the quality and emotional value were more important in explaining perceptions. However, all four value dimensions had a significant influence on the three outcome variables. Indeed, using stepwise regression, each dimension entered the equation separately and significantly. This indicates that each value dimension plays an important and separate role in forming attitudes and behaviors in the purchase process.

TABLE 6
Comparative Analysis of Models of Various Dimensionalities—Stage Three

Model	χ^2	DF	AGFI[a]	RNI[b]
a) Furniture store (after purchase)				
Null	3137.32	36	0.14	na.
One factor	814.63	27	0.51	0.67
Three factor	300.46	24	0.74	0.88
Four factor	56.38	21	0.92	0.99
b) Car stereo store (after purchase)				
Null	2308.29	36	0.11	n.a.
One factor	617.11	27	0.54	0.74
Three factor	194.11	24	0.78	0.93
Four factor	87.14	21	0.87	0.97

3.8. Data collection—stage 3

In the third stage, we evaluated the robustness of the 19-item scale in a different stage of the purchase process. Here, perceived value was evaluated *after* a purchase, rather than in the prepurchase situation studied in stages one and two.

Data were collected from customers of two different types of stores selling durable goods, a furniture store with two outlets and a car stereo center with three outlets. Customers were invited to participate in the survey immediately after committing themselves to a purchase. Only customers making a major purchase, $400 or more, were included. These customers were asked to take a self-completion questionnaire and return it in the reply paid envelope within ten days. Respondents were also offered the opportunity to participate in a drawing for a voucher redeemable at the store. The survey was conducted over a six-month time period and all qualifying customers were asked to participate.

Usable responses represented 44% of furniture store customers and 31% of car stereo center customers, providing samples of 323 and 313 customers respectively. To check for nonresponse bias, we compared sample profiles with company records of all such customers over the same time period in terms of variables such as suburb of residence and expenditure levels. A χ^2 test found no significant differences. We concluded that the sample was representative of major purchasers during this time period.

In this third stage, the suggested four-factor model was again tested for validity and reliability. As before, the four-factor model was supported over the three-factor, one-factor and null, as can be seen in Table 6. The four-factor perceived value model was also supported in a postpurchase context.

The reliabilities of the factors and the total scale for both postpurchase data sets are shown in Table 7. The reliabilities range from 0.84 to 0.95. This represents an improvement in the reliability of scales for quality, price and social value, although the emotional value scale was slightly less in stage three than in either of the two earlier stages. The total scale reliability was 0.95 in both cases.

Discriminant validity was again supported in stage three. The highest correlations were again between quality and emotional factors. With a corresponding standard error of 0.03,

TABLE 7
Correlations between Constructs following Confirmatory Factor Analysis—Stage Three

	Quality	Emotional	Price	Social
Furniture store (after purchase)				
Quality	0.95			
Emotional	0.77 (0.03)	0.86		
Price	0.61 (0.04)	0.52 (0.050	0.90	
Social	0.28 (0.05)	0.35 (0.05)	0.31 (0.05)	0.92
Car stereo store (after purchase)				
Quality	0.94			
Emotional	0.80 0.03)	0.86		
Price	0.77 (0.03)	0.76 (0.04)	0.84	
Social	0.39 (0.05)	0.48 (0.05)	0.33 (0.06)	0.91

NB Standard errors appear in brackets, whille composite reliabilities appear on the diagonal
Reliability of linear composite of scale (19 items) = 0.95 (furniture store) and 0.95 (furniture store) and 0.95 (Car stereo store)

the confidence interval surrounding the correlation between constructs shows them to be distinct. The average variance extracted, employing Fornell and Larcker's (1981) test, ranged from 0.76 to 0.85 for the furniture store and 0.73 to 0.84 for the car stereo store. The square of the path between the constructs was a maximum of 0.56 and 0.61 respectively.

4. Conclusions

4.1. Theoretical and managerial implications

In this study, we extend our knowledge of perceive consumer value by developing and testing a parsimonious and practical four-dimensional scale of this construct. Unlike previous measures, our construct includes both utilitarian and hedonic components. The importance of this combination can be seen in a comment by MacKay (1999, p 182), who noted that a product's or a service's appeal is an "amalgam of rational and emotional factors" and that "emotions play a part in every purchase decision (but) . . . very few purchases are entirely emotional." Our study shows that multiple value dimensions explain consumer choice better, both statistically and qualitatively, than does a single 'value for money' item and should produce superior results when investigating consumption value.

The reliabilities, factor structure and validity tests indicate that the 19-item PERVAL scale and its four dimensions have sound and stable psychometric properties. The scale demonstrates that consumers assess products, not just in functional terms of expected performance, value for money and versatility; but also in terms of the enjoyment or pleasure derived from the product (emotional value) and the social consequences of what the product communicates to others (social value). Additionally, the scale was found to be reliable and valid in a post purchase situation, as well as in a prepurchase situation.

Recognition of the importance of the different dimensions of value should enable retail marketers to develop more sophisticated positioning strategies. Our results show the importance of emotional value on consumer's willingness to buy in the durable product category normally considered as functionally oriented. This has substantial implications for marketing strategy. For example, many retailers experienced a downturn in sales due to a loss of confidence among consumers and an associated focus on price during the early part of the 1990s (Age, 1993). However, research tells us that the broader concept of value is often more important to many consumers (Zeithaml, 1988; Tellis & Gaeth, 1990; Swait & Sweeney, 2000). The shift of promotional and retail sales orientation to the other dimensions of value may be particularly useful under such conditions.

Such conditions illustrate the opportunity for retail strategists to explore all dimensions of customer value before deciding on an appropriate market approach. Since previous research (Sweeney et al., 1999) has found that retail service quality plays a significant role in the creation of value perceptions, the present study also provides potential insights for the recruitment and training of retail staff. For example, phrases such as "it must make me want to buy it or use it" arose several times in the initial focus groups. If the front-line staffers have an understanding of the multidimensional nature of value, and its impact on consumer decision-making processes, they can build aggregate value without resorting to discounts. The ability of retail staff to build emotional and social values, and the knowledge as to when this is important, may pay substantial dividends.

The value of the brands it carries is a major asset for the retailer in its battle for increased customer loyalty and associated higher profits. Webster (2000) discusses the impact of brand on retailer outcomes and the importance of incorporating consumer perceived value with respect to the brand with a value proposition for the retailer. Successful retailers deliver genuine value to customers through their commitment to the products sold as well as their retailing concept (Berry, Seiders & Gresham, 1997). Such retail strategies enhance store-merchandising authenticity by acknowledging and blending the various dimensions of consumer perceived, product value into their market positions.

4.2. Limitations and future research

The extent to which our findings may be extended to all retail categories remains to be explored. As previously explained, we employed consumer reactions to durable goods in the later stages of our scale development. Whether the resulting scale will apply equally in other product contexts cannot be stated. However, we believe the majority of the scale is likely to remain appropriate for a variety of contexts with only quality items likely to need adaptation for nondurable products.

We also note that the epistemic and conditional values suggested by Sheth et al. (1991a) were not specifically tested in this study. Sheth et al. (1991a, p 69) defined conditional value as "derived from its capacity to provide temporary functional or social value in the context of a specific and transient set of circumstances or contingencies." While this must be empirical evaluated, our testing leads us to doubt that this conditional value component is of the same order as other dimensions. Conditional value is derived from the moderating effect

of a situation on perceptions of functional and social value on outcomes. Situational aspects, by themselves, may moderate the effect of emotional value on outcomes.

Nonetheless, we urge that epistemic value be considered in future research, particularly when considering products for which novelty or surprise are important. Rarity value may also need to be considered for certain products (e.g., prestigious European cars) as commodity theory claims, "any commodity will be valued to the extent that it is unavailable" (Brock, 1968).

The present study developed a perceived value scale in both pre and postpurchase situations. However, value assessment criteria may change across these scenarios and this needs evaluation (Woodruff, 1997; Parasuraman, 1997). For example, consequences may be more important attributes when determining value after the use of the product.

Another fruitful research direction is the study of the role that the feedback loop plays in the consumer decision-making model. This loop suggests that the experiences of a previous purchaser influence future perceptions. Current perceived value may be affected by previous experiences with the same product type (the repurchase of the product type), the same brand (brand loyalty) or even the same store (store loyalty). Perceived product value should be viewed as a part of a continuous process in the maintenance of a relationship between a manufacturer and retailer with a given customer. Studies employing a longitudinal framework may be productive may shed different light upon the relative importance, and perhaps the susceptibility of change of the four dimensions of value.

Notes

1. The results of the factor analysis on the data collected in the first stage are available on request from the authors.

Appendix

A partial-disaggregation approach was used in the various confirmatory factor analyses undertaken in the present study. This approach is a compromise between the most aggregative approach, in which all items are summed to form one composite for a construct, and the most disaggregative, in which each item is treated as an individual indicator of the relevant factor (Bagozzi & Heatherton, 1994; Bagozzi & Foxall, 1996). The former approach suffers from a loss of information, as the distinction among items is lost. The latter is unwieldy due to high levels of random error in items and the need for many parameters to be estimated (Bagozzi & Heatherton, 1994; Bagozzi & Foxall, 1996).

The partial-disaggregation approach allows the researcher to combine items into composites to reduce random error, yet retain the multiple measure approach to structural equation modeling. In practice, the items that load on a factor under exploratory factor analysis are selected at random to form two or three combined indicators rather than using single item measures. Since all indicators of a construct should correspond in the same way to that latent variable, different random combinations should lead to the same fit.

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